

RAMESH REDDY CHANGAL

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Software Engineer · Applied AI/ML · Backend Systems · Python

Technical Skills

AI/ML: LangChain, RAG Pipelines, Hugging Face Transformers, PyTorch, Scikit-learn, Stable Diffusion, LLMs (Mixtral, GPT-4), OpenAI API, Anthropic API, NumPy, Pandas
Languages: Python, JavaScript, TypeScript, Java, SQL
Backend & APIs: FastAPI, Flask, Spring Boot, Node.js, REST APIs, WebSockets, OAuth 2.0
Frontend: React, Next.js, TypeScript, Tailwind CSS
Infrastructure: Docker, Kubernetes, AWS (EC2, S3, Lambda), PostgreSQL, MongoDB, Redis, CI/CD, Git, Claude Code (daily use — code generation, refactoring, debugging, productivity acceleration)

Professional Experience

Machine Learning Research Intern

International Crops Research Institute for the Semi-Arid Tropics (ICRISAT)

Sep 2022 – Oct 2022

Hyderabad, India

- Enabled data-driven farming risk assessment for 3-4 research scientists by deploying ML crop yield prediction model achieving 70% accuracy using Linear Regression and Random Forest on on-premises server accessible via local WiFi.
- Improved data reliability by 40% through systematic cleaning of noisy agricultural datasets; built Streamlit interface eliminating API overhead, making predictions accessible to non-technical users; contributed to DHIS2 data collection tool with PostgreSQL integration.

Software Development Intern

Path Creators

Jun 2021 – Jul 2021

Hyderabad, India

- Enhanced user engagement by delivering pattern-based conversational chatbot with Python, Flask, and JavaScript; designed conversation flow logic and integrated backend-frontend systems for multi-turn interactions.

Software Development Intern

Path Creators

Jun 2019 – Jul 2019

Hyderabad, India

- Built real-time obstacle-detection IoT prototype using Arduino and sensor arrays; led hardware-software integration within a 4-member team and shipped functional system in 36 days against a tight client deadline.

Graduate Grading Assistant – Discrete Mathematics

University of Maryland, Baltimore County

Mar 2024 – May 2025

Baltimore, MD

- Supported 150+ students across 3 semesters by grading assignments and exams in probability theory, graph theory, and combinatorics; held office hours covering advanced topics when primary TA was unavailable.

Key Projects

MailMind – AI Gmail Intelligence Platform | FastAPI · LangChain · OAuth 2.0 · LLM · React

- Gmail's keyword search breaks down for context-aware queries across large inboxes; built a natural language search interface using LangChain to translate user queries into Gmail API filters — no email data stored on the backend.
- OAuth 2.0 with read-only scope enforces access boundaries at the auth layer; FastAPI backend handles async fetching with response caching to keep latency low across high-volume inboxes.
- Built an NLP-based analytics layer on top of email content — extracts transaction data, categorizes expenses, and tracks spending patterns without requiring users to manually export or tag anything.

Multi-Agent Research Platform | Python · Mixtral-8x7B · LangChain · BeautifulSoup · Hugging Face · arXiv API

- Built a two-stage LLM verification pipeline to address hallucination on post-training-cutoff topics — first Mixtral-8x7B instance retrieves and analyzes 15 live sources (10 web via DDGS + 5 arXiv papers), second instance synthesizes a grounded response from that analysis.
- Separated retrieval from generation by design: BeautifulSoup handles web extraction, arXiv API fetches recent papers — prevents retrieval noise from propagating into the final output, a known failure mode in single-pass RAG.
- Responses include source citations, making output auditable for topics like recent model releases or policy updates where a confident but wrong answer has real consequences.

Distributed File Storage System | Python · Flask · Docker

- Built a GFS-inspired distributed storage system — master node tracks chunk metadata and assignments, three storage nodes handle persistence; files chunked at 512 bytes and replicated across nodes so the system tolerates two simultaneous node failures.
- Clients fetch chunk-to-node assignments from master and write directly to storage nodes, keeping master out of the data path and preventing it from becoming a throughput bottleneck.
- Implemented Flask heartbeat monitoring with automatic re-replication on node failure; containerized with Docker, sustained 99.9% uptime in failure-injection testing; adding storage nodes requires no migration or downtime.

AI Icon Generation Platform | Python · FastAPI · React · LangChain · Llama · Stable Diffusion · Hugging Face

- Diffusion models are sensitive to prompt quality — vague inputs produce unusable output; built a prompt optimization pipeline where user input and style preferences are passed through LangChain to Llama, which rewrites them into a well-structured Stable Diffusion prompt.
- Stable Diffusion runs 4 inference passes on the refined prompt via Hugging Face; all 4 outputs are converted from raw tensor format to web-compatible images and returned together so users can compare results rather than commit to a single generation.
- FastAPI owns the full backend pipeline — prompt rewriting, inference, format conversion, response delivery; React frontend handles display and selection only, keeping all generation logic server-side.

Education

University of Maryland, Baltimore County (UMBC)

Master of Science in Computer Science – GPA: 3.6/4.0

Aug 2023 – May 2025

Baltimore, MD